Softmax from Fast Mixing

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August 2026 — working draft

Abstract

Softmax is used as a choice law almost everywhere, usually without a mechanism that produces it. We give one. When N channels compete under a shared hidden state that mixes faster than the channels fire, the winning probability is softmax in the stationary mean intensities at leading order, and the first correction is a Green–Kubo term built from integrated rate autocovariances. One pair of objects, the mean intensities and the correlation matrix K, then predicts every blocked-subset counterfactual by restricting sums, so the whole deletion ensemble follows from one calculation. Fluctuations common to all channels cancel exactly, so departures from softmax measure differential structure alone, and environmental modes of rank r give K of rank r, up to a cap set by the environment's own dimension. The same correction predicts the runner-up kernel, which is observable in any race that reports a finishing order, so the theory can be tested without running interventions. We verify the expansion on finite chains, then on Brownian narrow escape with reaction windows, where the leading and corrected errors decay with measured orders 0.93 and 1.93 against an exactly discretized generator.

1 Introduction

A choice among competing options is often modelled by softmax. The rule assigns option i a probability proportional to e^{θ_i}, and it appears in discrete choice, reinforcement learning, statistical physics, and the output layer of essentially every classifier. Its usual justification is convenience. Softmax is smooth, normalized, and differentiable, and in the Gumbel random-utility model it is exact.

Underneath many applications, though, there is real kinetics. A chemical reaction fires when one of several channels crosses a barrier first. A diffusing particle escapes through whichever window it encounters first. A kinetic Monte Carlo (KMC) step selects among escape channels by their rates. In these systems the choice is a race, and the race has a mechanism. It is worth asking what that mechanism implies about the choice law, rather than assuming softmax and moving on.

This paper answers that for one specific and common situation. The channels share a hidden state, and that state mixes fast relative to the rate at which channels fire. Catalysis supplies the picture. Adsorbate configurations rearrange quickly while the reaction event is rare, so every channel sees an environment that has effectively averaged out. Under that separation we show two things.

*Code and machine verification for every claim in this paper: `https://github.com/micropediction/kinetics` (experiments 7 and 9). Web companion: `https://kinetics.microprediction.org`.

The leading-order result is that softmax is exact in the limit, with $\theta_i = \log \bar{\lambda}_i$ the log of the stationary mean intensity. Fast mixing homogenizes the environment, each channel is left with its average rate, and the race among constant rates is the exponential race, whose choice law is Luce.

The first correction is a Green–Kubo term. It is built from $K_{jk} = \int_0^\infty \mathrm{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t))\, dt$, the time-integrated covariance of the firing intensities under the stationary law. This is a dynamical object rather than a static one. Two channels whose rates are correlated at equal times contribute nothing unless the correlation persists over the timescale on which the environment decorrelates.

The correction is what breaks independence of irrelevant alternatives (IIA). Softmax redistributes a removed channel's probability in proportion to the survivors' probabilities, which is the IIA prediction and is usually wrong in physical systems. The Green–Kubo term supplies the deviation, and it does so for every blocked subset at once.

That last point is the practical one. The pair $(\bar{\lambda}, K)$ is computed once. Any counterfactual, remove channel 3, block channels $\{2, 5, 7\}$, is answered by restricting the sums in the correction to the surviving set. No new solve is required per counterfactual. This is the statistical counterpart of a theme running through our related work, that one global computation can contain an entire leave-one-out ensemble.

Two structural facts sharpen the result. Fluctuations shared by all channels, $\lambda_i(y) = a_i c(y)$, cancel from the correction identically. We show this is not merely an asymptotic statement. A common fluctuation is a common time change, so Luce holds exactly at every degree of separation, which is the proportional-hazards normal form. Separately, if the intensity deviations are driven by r environmental modes then K has rank r, so the correction inherits any low-dimensional structure the physics has.

Section 2 sets up the shared-state race and the killed resolvent. Section 3 states and proves the expansion. Section 4 collects the three structural properties. Section 5 verifies everything on finite chains, where each quantity is an exact linear solve. Section 6 moves to continuous physics, a reflected Brownian motion in a disk with reaction windows, and reports the measured convergence orders together with two systematic errors found there. Section 7 shows the correction also predicts the runner-up kernel, which makes it testable against ranked data.

2 The shared-state race

Let Y_t be an ergodic Markov process on a state space with generator L/ε and invariant law π. The parameter ε is the ratio of the environment's mixing time to the timescale on which channels fire. Small ε means fast mixing.

Each channel $i \in \{1, \ldots, N\}$ fires at intensity $\lambda_i(Y_t) \geq 0$, conditionally Poisson given the path of Y. The first channel to fire wins. For an available set $A \subseteq \{1, \ldots, N\}$ write $\Lambda_A(y) = \sum_{j \in A} \lambda_j(y)$ for the total hazard, and write $\bar{\lambda}_i = \int \lambda_i \, d\pi$ for the stationary mean intensity.

Let $u_i^A(y)$ be the probability that channel i wins, started from $Y_0 = y$. Conditioning on the first instant and using the conditional Poisson structure gives

$$\left(\frac{L}{\varepsilon} - \Lambda_A \right) u_i^A = -\lambda_i, \qquad u_i^A = \left(\Lambda_A - \frac{L}{\varepsilon} \right)^{-1} \lambda_i. \tag{1}$$

This is a killed resolvent. The generator propagates the environment and the multiplication operator Λ_A kills paths at the total firing rate. On a finite state space it is a linear system, which is what makes the verification in Section 5 exact.

The observable of interest is the win probability from a stationary start,

$$p_i^A = \int u_i^A \, d\pi. \tag{2}$$

Summing (1) over $i \in A$ gives $\sum_{i\in A} u_i^A = 1$, so the p_i^A are a probability vector on A for every ε.

3 Homogenization and the Green–Kubo correction

Write $\tilde{\lambda}_i = \lambda_i - \bar{\lambda}_i$ for the fluctuation of the ith intensity, and Π for the projection $\Pi f = (\int f d\pi)\mathbf{1}$ onto constants. The deviation operator

$$D = (\Pi - L)^{-1}(I - \Pi) \tag{3}$$

inverts L on the mean-zero subspace, and $Dg = \int_0^\infty e^{tL} g \, dt$ for mean-zero g. Note the sign it carries. Differentiating that integral gives $LDg = -g$, so D is minus the pseudo-inverse of L, and the sign propagates into the proof below. The Green–Kubo matrix is

$$K_{jk} = \int_0^\infty \mathrm{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t)) \, dt = \int \tilde{\lambda}_j \, (D\tilde{\lambda}_k) \, d\pi. \tag{4}$$

The second expression is what one computes. It replaces a time integral by a single linear solve. Note that K need not be symmetric. It is symmetric when the environment is reversible, and the verification in Section 5 deliberately uses a non-reversible chain so that the index placement in the theorem is actually tested.

Theorem 1 (Homogenized choice law and its first correction)**.** *Let $\bar{\Lambda}_A = \sum_{j\in A} \bar{\lambda}_j$ and suppose $\bar{\Lambda}_A > 0$. Then as $\varepsilon \to 0$,*

$$p_i^A = \frac{\bar{\lambda}_i}{\bar{\Lambda}_A} - \frac{\varepsilon}{\bar{\Lambda}_A}\left[\sum_{j\in A} K_{ji} - \frac{\bar{\lambda}_i}{\bar{\Lambda}_A}\sum_{j,k\in A} K_{jk}\right] + O(\varepsilon^2). \tag{5}$$

In particular $p_i^A \to \bar{\lambda}_i/\bar{\Lambda}_A$, the softmax law in the log mean intensities $\theta_i = \log \bar{\lambda}_i$.

Proof. Multiply (1) by ε to get $(L - \varepsilon\Lambda_A)u_i^A = -\varepsilon\lambda_i$, expand $u_i^A = u_i^{(0)} + \varepsilon u_i^{(1)} + \varepsilon^2 u_i^{(2)} + O(\varepsilon^3)$, and collect powers of ε.

At order ε^0, $Lu_i^{(0)} = 0$, so $u_i^{(0)}$ is a constant c_i by ergodicity.

At order ε^1 the equation reads $Lu_i^{(1)} = \Lambda_A c_i - \lambda_i$. A solution exists only if the right side is orthogonal to π, which forces $\bar{\Lambda}_A c_i = \bar{\lambda}_i$ and gives the leading term. The right side is then $c_i\tilde{\Lambda}_A - \tilde{\lambda}_i$ with $\tilde{\Lambda}_A = \sum_{j\in A}\tilde{\lambda}_j$, and inverting with $LD = -I$ on mean-zero functions gives

$$u_i^{(1)} = -D\Big(c_i\tilde{\Lambda}_A - \tilde{\lambda}_i\Big) + m_i,$$

where the constant $m_i = \int u_i^{(1)} d\pi$ is fixed at the next order.

At order ε^2 the equation is $Lu_i^{(2)} = \Lambda_A u_i^{(1)}$, whose solvability condition is $\int \Lambda_A u_i^{(1)} \, d\pi = 0$. The constant part of Λ_A contributes $\bar{\Lambda}_A m_i$, the constant m_i integrates against $\tilde{\Lambda}_A$ to zero, and what remains is

$$0 = \bar{\Lambda}_A m_i - \int \tilde{\Lambda}_A \, D\Big(c_i\tilde{\Lambda}_A - \tilde{\lambda}_i\Big) \, d\pi.$$

Using (4), $\int \tilde{\Lambda}_A D\tilde{\lambda}_i \, d\pi = \sum_{j\in A} K_{ji}$ and $\int \tilde{\Lambda}_A D\tilde{\Lambda}_A \, d\pi = \sum_{j,k\in A} K_{jk}$, so

$$m_i = -\frac{1}{\bar{\Lambda}_A}\left[\sum_{j\in A} K_{ji} - \frac{\bar{\lambda}_i}{\bar{\Lambda}_A}\sum_{j,k\in A} K_{jk}\right].$$

Finally $p_i^A = c_i + \varepsilon m_i + O(\varepsilon^2)$ by (2), which is (5). □

Remark 1. The two terms in the bracket have distinct roles. The first, $\sum_{j\in A} K_{ji}$, measures how channel i's rate correlates dynamically with the total hazard it competes against. The second subtracts the part of that effect which is common to all channels, and is what keeps $\sum_{i\in A} p_i^A = 1$ at each order.

4 Three structural properties

One computation answers every counterfactual. The pair $(\bar{\lambda}, K)$ does not depend on A. Every blocked-set counterfactual is obtained from (5) by restricting the sums to the surviving set. For a system with N channels this reduces 2^N separate resolvent solves to one deviation solve and $O(N^2)$ stored numbers, at the cost of the $O(\varepsilon^2)$ error the expansion carries.

Common-mode fluctuations cancel. Suppose $\lambda_i(y) = a_i c(y)$, so that every channel is modulated by the same environmental factor. Then $\tilde{\lambda}_i = a_i\tilde{c}$ and $K_{jk} = a_j a_k \gamma$ with $\gamma = \int \tilde{c}\,(D\tilde{c})\,d\pi$. Substituting into the bracket in (5) gives $a_i\gamma\sum_{j\in A} a_j - (a_i/\sum_{j\in A} a_j)\gamma(\sum_{j\in A} a_j)^2 = 0$. The correction vanishes identically.

Proposition 1 (Proportional hazards give Luce exactly)**.** *If $\lambda_i(y) = a_i c(y)$ with $c > 0$ then $p_i^A = a_i/\sum_{j\in A} a_j$ exactly, for every ε and every A, not merely to the order computed above.*

Proof. Let $\tau_c(t) = \int_0^t c(Y_s)\,ds$, a strictly increasing time change. In the clock τ_c every channel fires at the constant rate a_i, because the cumulative hazard of channel i up to real time t is $a_i\tau_c(t)$. The race is then a race among constant exponential rates, whose winner is i with probability $a_i/\sum_{j\in A} a_j$, independently of the clock. Since the identity of the winner is invariant under a common monotone time change, the same holds in real time. □

This is a stronger statement than the vanishing of the correction, and it is practically useful in the opposite direction. It says a measured departure from softmax cannot be produced by any amount of shared environmental fluctuation. Whatever deviation is observed is evidence of differential structure among the channels.

Low-rank environments give low-rank corrections. Suppose the intensity deviations are driven by r environmental modes, $\lambda_i - \bar{\lambda}_i = b_i^\top z(y)$ with z taking values in $\mathbb{R}^r$ and $\int z\,d\pi = 0$. Then $K = B\Gamma B^\top$ with B the $N \times r$ matrix of loadings and $\Gamma_{ab} = \int z_a(Dz_b)d\pi$, so K has rank at most r. There is a second cap. Deviations live in the mean-zero subspace of the environment, so $\operatorname{rank}(K) \le \min(N, m-1)$ for an environment with m states, whatever the loadings are. Experiment 40 attains both bounds, giving rank exactly r for $r = 1, \dots, 5$ on a six-state environment and rank $5 = \min(10, 5)$ for generic intensities on ten channels. The compressed correlated race that appears elsewhere in this program therefore has a physical provenance rather than being a modelling convenience.

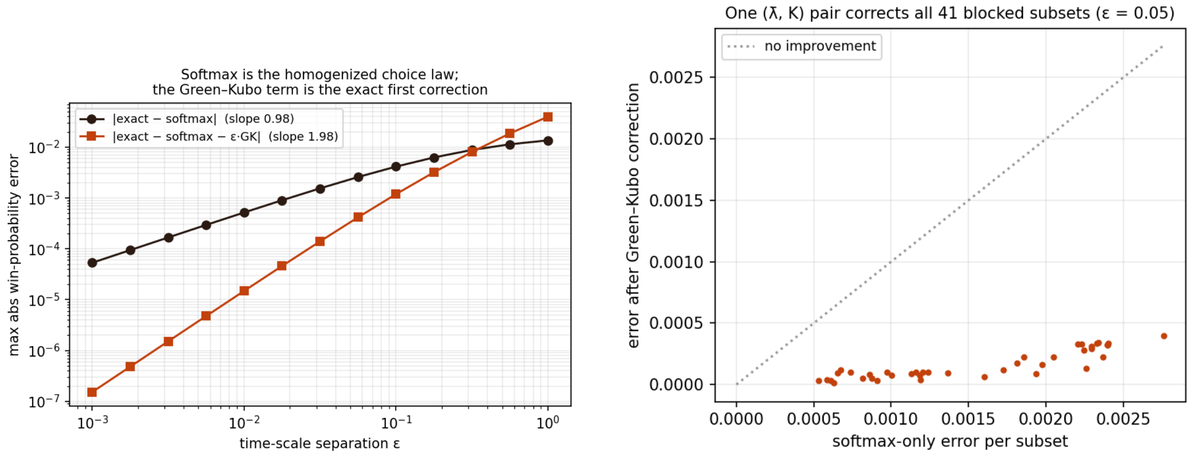

Figure 1: Finite-chain verification (experiment 7). Left, error against ε for the leading softmax law and for the Green–Kubo corrected law, with measured orders 0.985 and 1.985. Right, the same $(\bar{\lambda}, K)$ pair applied to 41 random availability sets at $\varepsilon = 0.05$, with no per-subset computation.

5 Verification on finite chains

On a finite state space every object above is an exact linear solve, so the theorem can be tested without simulation error. Experiment 7 runs a six-state chain with ten channels.

Setup. The generator has independent uniform off-diagonal rates on $[0.2, 1.5]$, giving a non-reversible chain with spectral gap 2.63. Channel intensities are lognormal with log-scale 0.8, giving $\bar{\Lambda}_A = 11.4$ on the full set. The seed is 5 and the sweep covers thirteen values of ε from 10^{-3} to 1. The dimensionless small parameter is $\varepsilon\bar{\Lambda}_A/\text{gap} \approx 4.3\,\varepsilon$, so the fitted window ends near 0.08 in those units and $\varepsilon = 0.3$ corresponds to 1.3.

The measured convergence order of the leading term is 0.985, against the theoretical 1. After adding the Green–Kubo correction the measured order is 1.985, against the theoretical 2. Both are fitted over the smallest six of thirteen values of ε, spanning 10^{-3} to 1.8×10^{-2}, since the expansion is asymptotic and the largest ε carry the $O(\varepsilon^2)$ residual. The gain of exactly one order is the signature the expansion predicts.

One instance is an anecdote, so experiment 40 repeats the measurement over twenty environments drawn with four to nine hidden states, three to eight channels, and intensity dispersions from 0.3 to 1.2. The leading order has median 0.960 and range $[0.919, 0.982]$, the corrected order has median 1.958 and range $[1.888, 1.979]$, and the gain exceeds 0.948 in every single trial.

Figure 1 shows both. The subset-uniformity claim is tested by fixing one $(\bar{\lambda}, K)$ pair and predicting 41 random availability sets at $\varepsilon = 0.05$. The maximum error over all subsets falls from 2.8×10^{-3} under softmax to 4.0×10^{-4} with the correction, with no per-subset computation.

The word Green–Kubo is doing work in (4), and experiment 40 tests whether it needs to. Replacing K by the equal-time covariance $\int \tilde{\lambda}_j \tilde{\lambda}_k \, d\pi$ scaled by a single timescale, with that timescale fitted to give the ablation its best case, leaves the measured order at 1.09 rather than 1.96, and the error at the smallest ε is 8.8 times worse. Static correlation between two channels does not determine the correction. The time integral does.

Common-mode cancellation is confirmed at 2.6×10^{-16} for the Green–Kubo coefficient, and the exact win probabilities agree with Luce to 1.1×10^{-16} at $\varepsilon = 0.3$. The regression suite carries the same check at $\varepsilon = 3.0$, where the environment is ten times slower than the channels rather than faster and the expansion has no claim at all, and Luce still holds to 10^{-12}. That is the content of Proposition 1. Rank-two environmental loadings give K with singular values $[0.022, 0.002, 0, 0]$, so the rank is exactly two.

A Gillespie simulation of the same system, event-driven and including the hidden state, agrees with the resolvent solution to 1.9×10^{-3} across 40,000 trajectories at $\varepsilon = 0.3$, against a sampling noise of about 2.7×10^{-3}.

The chain is not reversible, so K is asymmetric. On the committed instance $\max_{jk} |K_{jk} - K_{kj}| = 0.139$ against $\max_{jk} |K_{jk}| = 0.603$, a relative asymmetry of 23%. This is what distinguishes K_{ji} from K_{ij} in (5), and experiment 40 confirms the index order is not a convention. Transposing K in the correction degrades the measured order from 1.94 to 1.58, and K agrees with brute-force time integration of the covariance to 4.1×10^{-6}.

6 Narrow escape

Finite chains verify the algebra. To test the theorem as a statement about physics we take the hidden state to be reflected Brownian motion in the unit disk and the channels to be reaction windows on the boundary, bands in which channel i fires at rate κ. The first window to fire wins. Here κ plays the role of ε, since a slow reaction rate means the particle mixes many times before firing.

Setup. Sixteen boundary windows of half-width 0.12 on the unit circle, drawn with seed 42, three pairs of which overlap. The polar grid is 40×88, giving 3520 cells. Simulations use time step 2×10^{-4} and 50,000 trajectories per κ, with $\kappa \in \{2, 5, 10, 20, 40\}$.

Experiment 9 separates three layers so that no error source can hide in another. Layer A is a real Brownian simulation with independent per-step firing. Layer B is an exact killed-resolvent solve on a finite-volume Neumann Laplacian over a polar grid, with the uniform invariant law, reversibility, and conservation checked numerically. Layer C is the theory, band-area softmax plus the Green–Kubo term from one deviation solve, using the same code that passed the chain tests.

Comparing layers B and C tests the theorem on a continuous-physics generator. The softmax error decays with measured order 0.93 and the corrected error with order 1.93, again a clean gain of one order. Both orders are fitted on the three smallest values of κ, for the same reason as before.

Table 1 gives the full sweep, including where the correction stops paying. The improvement factor falls monotonically from $31\times$ at $\kappa = 2$ to $1.6\times$ at $\kappa = 40$. This is the expected behaviour of a first-order expansion rather than a defect, but it bounds the useful range, and a practitioner should read the correction as buying roughly an order of magnitude only while κ stays in the first half of this table.

Figure 2 shows the three layers together. Comparing layers A and B tests the discretization. The maximum discrepancy is 2.2×10^{-3} at $\kappa = 5$ with 50,000 trajectories against a sampling noise of about 10^{-3}, and falls to 9.6×10^{-4} against a noise floor of 7×10^{-4} in a 200,000-trajectory check.

κ	softmax	with Green–Kubo	improvement
2	1.6×10^{-3}	5.0×10^{-5}	$31\times$
5	3.7×10^{-3}	3.0×10^{-4}	$12\times$
10	6.9×10^{-3}	1.1×10^{-3}	$6.3\times$
20	1.2×10^{-2}	3.9×10^{-3}	$3.2\times$
40	2.0×10^{-2}	1.2×10^{-2}	$1.6\times$

Table 1: Maximum absolute error in the win probabilities against the exactly discretized generator, over sixteen boundary windows. The correction gains an order of magnitude at small κ and degrades to nothing by $\kappa = 40$, which is where the neglected $O(\kappa^2)$ term takes over.

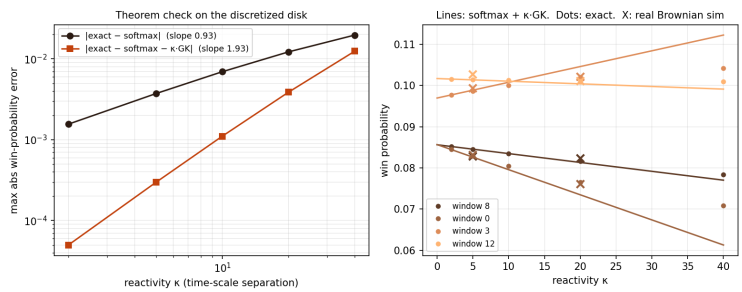

Figure 2: The theorem on continuous physics (experiment 9). Reflected Brownian motion in the unit disk with boundary reaction windows, error against κ for the leading and corrected laws, measured orders 0.93 and 1.93 against an exactly discretized Neumann generator.

Two systematic errors. Both would have corrupted a direct comparison of simulation against theory, and neither is visible without the middle layer.

The first is band quantization, worth 5×10^{-2}. Sharp cell indicators mis-size any window narrower than a grid cell, by up to a factor of three. Fractional area-exact cell coverage fixes it, after which band areas match their analytic values to 10^{-14}.

The second is overlap semantics, worth 1.4×10^{-2}. The window geometry contains three overlapping pairs. The original simulation awarded an overlap to the lower-indexed window, while the model stacks intensities as $\lambda = \kappa \sum_i g_i$, which is the physically correct convention for independent receptors. The error's structure identified it, being -0.014 on one window and $+0.006$ on its overlap partner, and its flatness under both time-step and grid refinement confirmed that it was not a discretization artifact.

7 The correction predicts the runner-up

The correction has been described so far as a statement about win probabilities, which are what a blocked-set experiment measures. Those experiments are expensive, since each availability set requires its own intervention. There is a cheaper observable that carries the same information.

Let $M_{ij} = \mathbb{P}(\text{runner-up} = j \mid \text{winner} = i)$ be the runner-up kernel, the probability that j finishes second given that i finishes first. Removing channel i changes the outcome only in those realizations where i won, and in those the runner-up inherits the win. This gives an identity that holds for every ε and involves no approximation,

$$q_j^{(-i)} = p_j + p_i M_{ij}, \tag{6}$$

where $q^{(-i)}$ is the deletion counterfactual. The companion work shows M is exactly the object that identifies single deletions, and that winner-only data constrains it not at all.

Applying Theorem 1 to the full set and to $A \setminus \{i\}$ therefore predicts M itself. At leading order the prediction is

$$M_{ij} = \frac{\bar{\lambda}_j}{\bar{\Lambda}_A - \bar{\lambda}_i} + O(\varepsilon), \tag{7}$$

which says the runner-up is drawn from the race among the remaining channels, the IIA prediction. Substituting the corrected shares into (6) gains an order.

Experiment 39 verifies this on the same chains. The identity (6) holds to 1.8×10^{-15}, the predicted kernel is row-stochastic to 8.0×10^{-15}, and the measured convergence orders are 0.967 for the Luce prediction (7) and 1.967 with the Green–Kubo correction. At $\varepsilon = 0.005$ the maximum error over off-diagonal entries falls from 3.0×10^{-4} to 2.1×10^{-6}.

This matters for testing the theory. The runner-up is observed in any race that reports a finishing order, so the correction can be checked against ordinary ranked data rather than against interventions. It also runs the other way. Where the environment is unknown, a measured M constrains the Green–Kubo matrix, which is the beginning of estimating K from data rather than computing it from a known generator.

8 Reproducibility

Every number in this paper comes from a committed script. Experiment 7 gives the chain verification, experiment 9 the narrow-escape study, experiment 39 the runner-up bridge, and experiment 40 the replication, the equal-time ablation, the rank bounds, and the index-order check. All four run from a clean checkout of `https://github.com/microprediction/kinetics` with NumPy, SciPy, and Matplotlib, and each writes the `results.csv` the corresponding numbers are quoted from. Experiments 7, 39, and 40 take seconds. Experiment 9 takes about three minutes. A regression suite locks every property stated here, including the two convergence orders, the exactness of common-mode cancellation, the rank bounds, and the runner-up identity.

9 Discussion

The result explains why softmax is a reasonable default in systems with fast shared dynamics, and it quantifies exactly when that default fails. The failure is governed by integrated dynamical correlations, not by static ones, and it is insensitive to any fluctuation shared across channels.

Four limitations bound the claim. The expansion is in the mixing-to-firing timescale ratio and says nothing when channels fire faster than the environment relaxes. The correction is first order, so systems at moderate separation retain an $O(\varepsilon^2)$ residual, quantified in Table 1, where the improvement decays to nothing by $\kappa = 40$. The theorem is proved for finite chains, and a formal treatment of reversible diffusions with a spectral gap is left open, so the narrow-escape study verifies rather than proves that case. Finally the continuum study is one geometry with one window arrangement, and while the chain results replicate over twenty environments, the physics results do not yet replicate over geometries.

One extension looks tractable. The disk's boundary operator diagonalizes in Fourier modes, which may render the leading geometric correction analytic rather than numerical.

Section 7 closes the gap between the computed correction and the estimated one, at least in one direction. What remains is the inverse problem. Estimating K from an observed runner-up kernel is an underdetermined fit, since M supplies $N(N-1)$ numbers for the N^2 entries of K and only through the combinations appearing in (5), so the identifiable part of K has yet to be characterized.